Dear Professor Abdelnabi,

I am writing to apply for the fully funded PhD position on Contextual Integrity for Autonomous AI Agents. I am currently pursuing an M.Sc. in Biomedical Engineering at the University of Tehran, where my research focuses on AI-enhanced optical imaging and the characterization of biological soft tissues under photothermal stimulation. With a B.Sc. in Mechanical Engineering and research experience spanning machine learning, optical imaging, signal processing, and robotics, I have developed an interdisciplinary background that I believe provides a strong foundation for studying reliable and responsible autonomous AI systems.

My current graduate research has given me hands-on experience at the intersection of machine learning and experimental scientific systems. As a Graduate Research Assistant, I have been developing an optical-based AI-enhanced imaging system, with particular emphasis on Generative Adversarial Network methods for image generation and enhancement. My broader thesis work involves developing and experimentally characterizing photothermal optical imaging systems and investigating the relationship between optical measurements and the mechanical behavior of biological tissues. This work has required me to design experiments, process and analyze complex data, investigate methodological limitations, and iteratively improve an experimental system based on observed failure modes.

My previous research experience has also exposed me to autonomous and robotic systems. At the University of Regina, I worked on developing a torque controller for a rope-climbing robot, where I investigated relevant control approaches and communicated research progress on a regular basis. I have additionally developed projects involving ROS/ROS 2, robot navigation, path planning, and an ESP32-based smart blind-assistance system. These experiences have made me particularly interested in how intelligent systems make decisions while interacting with uncertain environments and other agents.

The research questions in this project strongly resonate with the direction in which I would like to develop as a researcher. In particular, I am fascinated by the problem of how an autonomous agent should determine not only whether information is available, but whether it is appropriate to use or disclose in a particular context. Current AI systems can combine instructions, memories, retrieved information, and tool outputs remarkably effectively, yet this same ability can create serious problems when information is stale, manipulated, obtained under different assumptions, or outside the authority granted by a user. I find the distinction between information that an agent can access and information that an agent should use especially compelling.

Among the proposed directions, I am particularly interested in dynamic benchmarks, long-horizon evaluation, and adversarial/context-poisoning scenarios. My experimental background has taught me that reliable evaluation requires carefully controlled conditions, explicit failure analysis, and consideration of how errors accumulate through a system rather than focusing only on its final output. I would be excited to apply this perspective to autonomous agents, for example, by investigating how changing permissions, relationships, memories, and environmental conditions affect an agent's privacy decisions over long horizons.

I am also interested in exploring the intersection between contextual integrity and multi-agent systems. My background in robotics and control has given me an appreciation for systems in which multiple components must coordinate while operating under different constraints. Extending this perspective to agents representing different users raises fundamental questions about trust, authority, information sharing, and emergent privacy failures. I would be particularly interested in developing empirical environments in which these factors can be systematically varied and evaluated.

Technically, I bring experience with Python, MATLAB, C/C++, Git, machine learning, GAN-based methods, signal and image processing, robotics, and ROS/ROS 2. I have completed formal machine-learning training and have experience independently learning new technical areas to support research projects. My publication record includes a first-author conference paper on super-resolution GANs for photothermal OCT signal enhancement, as well as a journal manuscript currently under review.

My research path has consistently involved moving across disciplinary boundaries, from mechanical engineering and robotics to biomedical imaging and machine learning. I am particularly motivated by research questions that require both rigorous technical investigation and consideration of how intelligent systems should behave in real-world contexts. I am confident in my ability to develop the additional theoretical and practical expertise required for privacy, contextual integrity, agent evaluation, and trustworthy AI research.

I would be very excited to contribute to your work at the ELLIS Institute Tübingen and the Max Planck Institute for Intelligent Systems, and to develop research at the intersection of autonomous agents, privacy, reasoning, and responsible AI.

Thank you for considering my application. I would be grateful for the opportunity to discuss how my background and research interests could contribute to this project.

Sincerely,

Amir Hossein Osooli
M.Sc. Student in Biomedical Engineering
University of Tehran